

alcohols compared with alkanes (**see also 2.2.2 l and 4.1.2 c**)

- (ii) classification of alcohols into primary, secondary and tertiary alcohols

Reactions of alcohols

- (b) combustion of alcohols

- (c) oxidation of alcohols by an oxidising agent, e.g. $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ (i.e. $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$), including:
 - (i) the oxidation of primary alcohols to form aldehydes and carboxylic acids; the control of the oxidation product using different reaction conditions
 - (ii) the oxidation of secondary alcohols to form ketones
 - (iii) the resistance to oxidation of tertiary alcohols
- (d) elimination of H_2O from alcohols in the presence of an acid catalyst (e.g. H_3PO_4 or H_2SO_4) and heat to form alkenes
- (e) substitution with halide ions in the presence of acid (e.g. $\text{NaBr}/\text{H}_2\text{SO}_4$) to form haloalkanes.

Equations should use [O] to represent the oxidising agent.

PAG7 (see also 6.3.1 c)

Mechanism **not** required.

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